Development of a Defect Estimation Method for CFRP Interstage Structures with Holes using Finite Element Analysis and Graph Neural Networks

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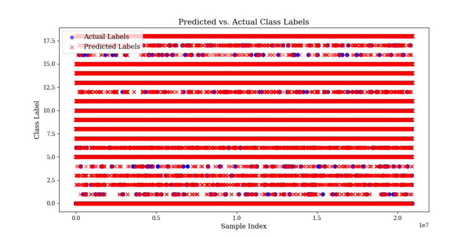
Motivation: defect localization in CFRP interstage structures

CFRP carries primary load in reusable space transportation (high strength-to-weight). **Interlaminar delamination** around bolt holes critically degrades integrity.

Infrared stress measurement (NDT) gives a full-field surface map of the *sum of principal stresses* (DSPSS) [1, 2]; sub-surface defects are inferred from it.



Predicted vs. true node labels on the perforated specimen.

Task

Per-node classification into 19 classes — *in-plane defect region* × *insertion layer* — directly on the FEA surface mesh.

Prior work and scope of this talk

- **Prior framework** (Frontiers in Materials, 2025) [3]: a GAT-based GNN localizes defects in perforated CFRP from FEA-simulated DSPSS, with *difference-normalized* inputs to suppress hole stress concentration.
- **This talk** extends it along two axes:
  1. robustness under *experimentally-motivated noise* + defect-free negatives;
  2. a controlled **architecture study** — from attention GNNs to *mesh-physics* GNNs and four novel MeshGraphNet variants.

Data generation

Synthetic FEA under tensile loading; interlaminar delamination modeled explicitly by **Teflon inserts** between selected plies [3].

Positioning within MS090E

This minisymposium is dominated by *PDE-solver / structure-preserving* methodology. Our contribution is the only **real-structure NDT pipeline**:

- vs. **thermodynamics-preserving GNNs** (Cueto et al.) [4, 5] — they solve/constrain physics; we *apply* GNNs to defect inference on real geometry.
- vs. **momentum-conserving physics-informed GNNs** (Sharma & Fink) — dynamics/conservation; we address static thermoelastic DSPSS.
- vs. **Fourier-embedded PINN** $\times$ **DIC** (Liu & Yi) [6] — full-field experimental measurement; shared sim-to-real motivation.

Three differentiators

(1) real engineering structure & NDT pipeline; (2) extreme class imbalance (defects $< 3\%$); (3) baseline-free & OOD generalization to unseen defect sizes.

Data & graph representation

One specimen → one graph

- 13,942 nodes (two stacked plies around the hole)
- Node features: $[x, y, z, \mathrm{DSPSS}]$
- Edges: FEA mesh connectivity, with coordinate-difference edge features $[dx, dy, dz, \|\cdot\|]$
- 19 classes (region × layer); class 0 = defect-free

Input = *difference-normalized* DSPSS (damaged − pristine) [3], isolating the defect signal from hole stress concentration.

The imbalance trap

Class-0 prior ≈ 0.998. Predicting "defect-free" everywhere gives 99.8% accuracy yet macro-F1 ≈ 0.

$\Rightarrow$ report **macro-F1**; engineer the training distribution, not only the loss.

Handling extreme class imbalance

The recipe that makes learning possible:

1. **Noise-augmented difference data** — robustness to acquisition noise [7].
2. **Defect-free negatives (NDF)** — 5000 genuine class-0 graphs (train only).
3. **Balanced sampling** 5000:5000 + **Focal loss** [8] (cf. LDAM [9], logit adjustment [10], GraphSMOTE [11]).
4. Leakage control: group-purged val/test split by specimen.

Dominant factor

With the recipe, a faithful reproduction reaches macro-F1 $\approx$ **0.70**, vs. ≤ 0.08 without it — the *data distribution* dominates.

Model zoo: from attention to mesh-physics GNNs

Model	Mechanism
GAT [12]	attention over neighbours (Frontiers baseline).
GATv2 [13]	*dynamic* attention (strictly more expressive).
GraphSAGE	max-aggregation of neighbour features.
PNA	multiple aggregators + degree scalers.
MeshGraphNet [14]	Encode-Process-Decode; node *and* edge updates each step [15].

All use coordinate-based geometric features (NDT-admissible). Equivariant GNNs [16] considered as future structure-aware option.

Architecture sweep under the balanced recipe

Identical recipe, 120 epochs, balanced 5000:5000, group-purged eval.

Architecture	macro-F1	note
MeshGraphNet [14]	**0.764**	Encode-Process-Decode, edge update
GATv2 [13]	0.694	dynamic attention
GAT [12]	0.674	Frontiers baseline (~0.61 published)
GraphSAGE	0.663	max-aggregation
PNA	0.613	multi-aggregator

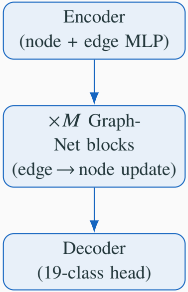
Takeaway

Under the balanced recipe, the **mesh-physics architecture** wins — beating the legacy GAT recipe (0.70) and the published baseline (0.61).

Why MeshGraphNet fits the problem

Encode–Process–Decode [14, 15]:

- node *and* edge embeddings updated at every message-passing step (residual);
- edge feature = coordinate difference $[dx, dy, dz, \text{dist}] \Rightarrow$ local geometry & anisotropy;
- matches FEA-mesh stress-propagation structure better than attention-only GNNs.



Novelty: four MeshGraphNet variants for layered NDT

Variant	Idea / why it could matter for CFRP
MeshGraphKAN	Fourier-KAN node encoder — learnable spectral basis for the *high-frequency stress-concentration* field (spectral-bias mitigation) [6].
Hybrid-MGN	mesh edges + *front/back cross-layer "world" edges* — structurally separates the two plies.
BiStride-MGN	per-graph global-context broadcast — multiscale, long-range stress propagation.
MGN-Transformer	interleaved per-graph self-attention — global node interaction.

Hypothesis

MeshGraphKAN × **stress concentration** and **Hybrid-MGN** × **two-ply separation** are the two novel angles most likely to cross macro-F1 0.80.

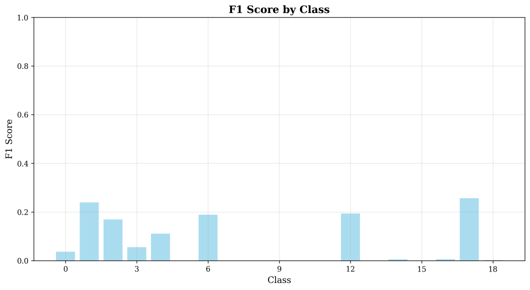
Variant results [runs in progress]

Variant	macro-F1	vs. MeshGraphNet (0.764)
MeshGraphNet (baseline)	0.764	—
MeshGraphKAN	TBD	
Hybrid-MGN	TBD	
BiStride-MGN	TBD	
MGN-Transformer	TBD	

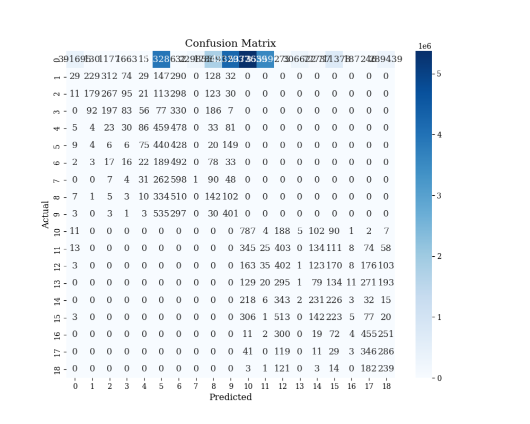
Target

Cross the **0.80** macro-F1 line with a physically-motivated variant.

Per-class performance & confusion



Per-class F1.



Confusion matrix (top classes).

Conclusions & outlook

- The **data recipe** (noise-aug difference + NDF negatives + balancing) is dominant: $0.08 \longrightarrow 0.70$.
- Under the balanced recipe, **MeshGraphNet** [14] is the best architecture (**0.764**), beating the published GAT baseline (0.61).
- Two **novel layered-NDT variants** (Fourier-KAN, front/back Hybrid) target > 0.80.

Future work

- **Sim-to-real:** transfer to experimental infrared-stress (TSA) data.
- **Physics consistency:** structure-preserving GNNs [4].
- **Application:** aerospace payload-fairing structural health monitoring.

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